1. Population and Sample

Population

Definition

A **population** is the complete set of all observations or data points that we want to study.

Simple Meaning

Population means **entire data group**.

Examples

- All students in a university
- All voters in a country
- All products produced in a factory
- All customers of an e-commerce website

Data Science Example

Suppose Netflix wants to analyze **user watching behavior**.

Population = All Netflix users worldwide.

Sample

Definition

A **sample** is a subset of the population selected for analysis.

Simple Meaning

Instead of studying entire population, we analyze **small portion of it**.

Example

Population = 10,000 students in university
Sample = 200 students selected for survey

Difference Between Population and Sample

Feature	Population	Sample
Definition	Entire dataset	Subset of dataset
Size	Very large	Smaller
Cost	Expensive	Cheap
Example	All voters	1000 voters surveyed

Why Sampling is Needed

Studying the entire population is often **impractical**.

Reasons

1️⃣ **Cost**

Collecting data from everyone is expensive.

Example
National census requires huge cost.

2 **Time**

Population study may take years.

Example
Election surveys.

3 **Feasibility**

Sometimes population cannot be measured completely.

Example
Quality testing where product is destroyed.

Data Science Example

Suppose Amazon has **100 million users**.

Instead of analyzing full data, they take **random sample of users** to study buying behavior.

---

2. Types of Sampling

Sampling methods are mainly divided into **two categories**:

1 Probability Sampling
2 Non-Probability Sampling

Probability Sampling

Every member of the population has **equal chance of selection**.

This method produces **unbiased results**.

---

1. Simple Random Sampling

Definition

Each element in the population has an equal probability of being selected.

Example

Selecting students using **lottery method**.

Population = 1000 students
Randomly choose 100 students.

Real Life Example

Online survey tools randomly selecting participants.

---

2. Systematic Sampling

Definition

Selecting every **kth element** from population.

Example

Population = 1000 people
Select every **10th person**.

Steps

1. Choose starting point randomly
2. Select every kth observation

Example
10th, 20th, 30th, 40th person.

---

3. Stratified Sampling

Definition

Population is divided into **homogeneous groups (strata)** and samples are taken from each group.

Example

Population: University students

Strata:

- Engineering
- Commerce
- Arts

Then sample from each group.

Advantage

Ensures **representation of all groups**.

---

4. Cluster Sampling

Definition

Population is divided into clusters, and **entire clusters are randomly selected**.

Example

City has 100 schools.

Randomly choose 10 schools and survey all students in those schools.

Difference Between Stratified and Cluster Sampling

Feature	Stratified	Cluster
Groups	Homogeneous	Heterogeneous
Sampling	From each group	Entire group selected
Example	Departments	Schools

Non-Probability Sampling

In this method **not every element has equal chance of selection**.

Examples

- Convenience sampling
- Judgment sampling
- Snowball sampling

Example
Surveying people available nearby.

---

3. Sampling Distribution

Definition

The probability distribution of a sample statistic (such as mean) obtained from multiple samples.

Simple Meaning

When we take **many samples from a population**, each sample has a mean.

The distribution of those sample means is called **sampling distribution**.

Example

Population marks average = 60

Samples:

Sample 1 mean = 58
Sample 2 mean = 61
Sample 3 mean = 59
Sample 4 mean = 62

If we plot all sample means → we get **sampling distribution**.

Important Point

Sampling distribution helps us understand **how sample statistics behave**.

4. Standard Error

Definition

Standard deviation of the sampling distribution.

Formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

Where

σ = population standard deviation
n = sample size

Meaning

Standard error tells us **how much sample mean differs from population mean**.

Large sample → smaller error.

5. Central Limit Theorem (CLT)

Definition

The Central Limit Theorem states that the sampling distribution of the sample mean approaches a **normal distribution** as the sample size becomes large (usually $n \geq 30$), regardless of the population distribution.

Simple Meaning

Even if population is not normal,

large samples will produce **normal distribution of sample means**.

Graph Explanation

Population Distribution

May look like anything

Irregular Shape

Sampling Distribution

Becomes normal

```
   /\
  /  \
 /    \
/      \
```

Why CLT is Important

CLT allows us to use **normal distribution methods** for inference.

Real Life Data Science Example

Suppose a company wants to know **average delivery time**.

Population = All deliveries.

Instead of analyzing millions of deliveries, they:

- take samples of deliveries
- compute sample mean
- apply CLT to estimate population mean.

- # Python Practical for Class
- ## Sampling Distribution Visualization

```
import numpy as np

import matplotlib.pyplot as plt


population = np.random.normal(50,10,10000)


sample_means = []


for i in range(1000):
    sample = np.random.choice(population,30)
    sample_means.append(np.mean(sample))
plt.hist(sample_means,bins=30)
plt.title("Sampling Distribution of Sample Mean")
plt.show()
```